

AA SL 2023 May P1 TZ1 (no calculator)

Full marks are not necessarily awarded for a correct answer with no working. Answers must be supported by working and/or explanations. Where an answer is incorrect, some marks may be given for a correct method, provided this is shown by written working. You are therefore advised to show all working.

Section A

Answer all questions. Answers must be written within the answer boxes provided. Working may be continued below the lines, if necessary.

1. [Maximum mark: 5]

Point P has coordinates $(-3, 2)$, and point Q has coordinates $(15, -8)$. Point M is the midpoint of $[PQ]$.

- (a) Find the coordinates of M . [2]

Line L is perpendicular to $[PQ]$ and passes through M .

- (b) Find the gradient of L . [2]

- (c) Hence, write down the equation of L . [1]

2. [Maximum mark: 7]

The function f is defined by

$$f(x) = \frac{7x + 7}{2x - 4}$$

for $x \in \mathbb{R}$, $x \neq 2$.

(a) Find the zero of $f(x)$. [2]

(b) For the graph of $y = f(x)$, write down the equation of

(i) the vertical asymptote; [1]

(ii) the horizontal asymptote. [1]

(c) Find $f^{-1}(x)$, the inverse function of $f(x)$. [3]

3. [Maximum mark: 6]

On a Monday at an amusement park, a sample of 40 visitors was randomly selected as they were leaving the park. They were asked how many times that day they had been on a ride called The Dragon. This information is summarized in the following frequency table.

Number of times on The Dragon	Frequency
0	6
1	16
2	13
3	2
4	3

It can be assumed that this sample is representative of all visitors to the park for the following day.

(a) For the following day, Tuesday, estimate

(i) the probability that a randomly selected visitor will ride The Dragon;

(ii) the expected number of times a visitor will ride The Dragon. [4]

It is known that 1000 visitors will attend the amusement park on Tuesday. The Dragon can carry a maximum of 10 people each time it runs.

(b) Estimate the minimum number of times The Dragon must run to satisfy demand. [2]

4. [Maximum mark: 6]

(a) Show that the equation $\cos 2x = \sin x$ can be written in the form $2 \sin^2 x + \sin x - 1 = 0$. [1]

(b) Hence, solve $\cos 2x = \sin x$, where $-\pi \leq x \leq \pi$. [5]

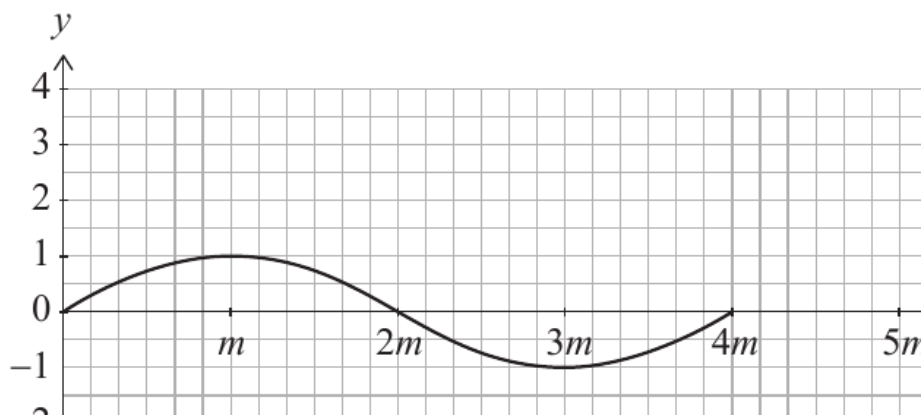
5. [Maximum mark: 6]

Find the range of possible values of k such that $e^{2x} + \ln k = 3e^x$ has at least one real solution.

6. [Maximum mark: 6]

The function f is defined by $f(x) = \sin qx$, where $q > 0$. The following diagram shows part of the graph of f for $0 \leq x \leq 4m$, where x is in radians. There are x -intercepts at $x = 0, 2m$ and $4m$.

The graph of f for $0 \leq x \leq 4m$, where x is in radians. There are x



(a) Find an expression for m in terms of q . [2]

The function g is defined by

$$g(x) = 3 \sin\left(\frac{2qx}{3}\right), \quad 0 \leq x \leq 6m.$$

(b) On the axes above, sketch the graph of g . [4]

Section B

Answer all questions in the answer booklet provided. Please start each question on a new page.

7. [Maximum mark: 13]

The function h is defined by $h(x) = 2xe^x + 3$, for $x \in \mathbb{R}$. The following diagram shows part of the graph of h , which has a local minimum at point A .

n h is defined by $h(x) = 2xe^x + 3$, for x
f h , which has a local minimum at poin



- (a) Find the value of the y -intercept. [2]
- (b) Find $h'(x)$. [2]
- (c) Hence, find the coordinates of A . [5]
- (d) (i) Show that $h''(x) = (2x + 4)e^x$. [2]
- (ii) Find the values of x for which the graph of h is concave-up. [2]

8. [Maximum mark: 14]

Consider the arithmetic sequence $u_1, u_2, u_3, \dots$

The sum of the first n terms of this sequence is given by $S_n = n^2 + 4n$.

(a) (i) Find the sum of the first five terms.

(ii) Given that $S_6 = 60$, find u_6 . [4]

(b) Find u_1 . [2]

(c) Hence or otherwise, write an expression for u_n in terms of n . [3]

Consider a geometric sequence, v_n , where $v_2 = u_1$ and $v_4 = u_6$.

(d) Find the possible values of the common ratio, r . [3]

(e) Given that $v_{99} < 0$, find v_5 . [2]

9. [Maximum mark: 17]

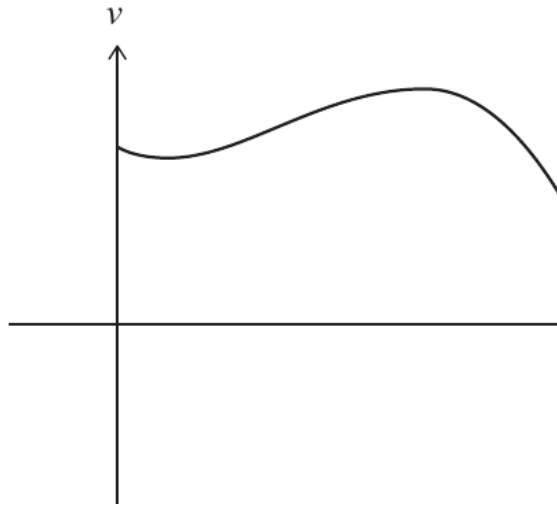
An object moves along a straight line. Its velocity, $v \text{ m s}^{-1}$, at time t seconds is given by

$$v(t) = -t^3 + \frac{7}{2}t^2 - 2t + 6, \quad 0 \leq t \leq 4.$$

The object first comes to rest at $t = k$.

The graph of v is shown in the following diagram.

h of v is shown in the following diagram.



At $t = 0$, the object is at the origin.

- (a) Find the displacement of the object from the origin at $t = 1$. [5]
- (b) Find an expression for the acceleration of the object. [2]
- (c) Hence, find the greatest speed reached by the object before it comes to rest. [5]
- (d) Find the greatest speed reached by the object for $0 \leq t \leq 4$. [2]
- (e) Write down an expression that represents the distance travelled by the object while its speed is increasing. Do not evaluate the expression. [3]